CMS L8 Test Protocol

Collision Management System – TORSA

EN-CMS-PROC-PRUEBAS-L8 · Version 1.0



20 October 2026

Record Tables

Change History

Version	Date	Name	Changes
1.0	2026-10-20	TORSA Engineering	Generic CMS L8 protocol, anonymised from a customer test protocol; L9 tests removed; tentative schedule added

Reviews

Version	Date	Name	Position
1.0	—	Pending	Pending

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Document	EN-CMS-PROC-PRUEBAS-L8
Version	1.0
Date	2026-10-20
Classification	Confidential · Minimum level L2
Applies to	Functional validation of CMS Level 8 (alert controls) on an operating mine

1. OBJECTIVE

Validate the performance of the TORSA Collision Management System (CMS) at EMESRT Level 8 in critical mining interaction scenarios, in a controlled and repeatable way. The tests check that the system detects the interaction, generates progressive visual and audible alerts to the operator, and notifies the monitoring platform.

The protocol is part of the formal validation process with the customer. It gives an objective and auditable basis for deciding whether the system can go live.

2. SCOPE

This procedure applies to:

- Vehicle-to-vehicle (V2V) interactions between instrumented vehicles on haul roads, curves and intersections
- The digital cleaning request and authorisation flow between an auxiliary vehicle and a shovel
- Vehicle-to-object (V2O) detection of a safety berm, only where the AI camera add-on is installed (optional)

Does not apply to:

- Level 9 intervention tests (shovel swing lock, truck braking). They are covered by a separate protocol (see ES-CMS-PROC-ISA-INTERVENCION)
- Interactions with non-instrumented vehicles or pedestrians

3. BEFORE STARTING

3.1 Key Definitions

Term	Definition
LO (Local Object)	Instrumented vehicle with the greater capacity to manoeuvre or the primary responsibility for mitigating the risk in an interaction
RO (Remote Object)	Vehicle involved in the interaction whose preventive capacity is secondary to the LO
V2V (Vehicle-to-Vehicle)	Interaction between two instrumented vehicles
V2O (Vehicle-to-Object)	Interaction between an instrumented vehicle and an object, including a dump edge or safety berm detected by cameras
Level 8 (EMESRT)	Alert controls: the system warns the operator and notifies the platform
Level 9 (EMESRT)	Intervention controls: the system acts on the vehicle to avoid or mitigate a collision
EMESRT	International reference organisation in mining safety that defines the hierarchy of controls for collision systems

3.2 Equipment and Materials

Item	Specification	Notes
Instrumented vehicles	Trucks, auxiliary vehicle (tractor) and shovel, as required by each scenario	CMS installed, commissioned and registered on the platform
Cones and markers	As shown in the layouts of Annexes A to G	Distances marked from 0 m
Communications	V2V communication between vehicles; platform connection	Verified before each scenario
Recording	Event logging active on the vehicles and on the platform	Needed for the evidence of each repetition
Radio	Link between TORSA supervisor, mine supervisor and operators	Test perimeter control

3.3 Roles and Responsibilities

Role	Responsibility
TORSA supervisor	Technical execution and documentary support
Mine supervisor	Validation in the field
Operator	Controlled driving
Safety	Authorisation and perimeter control
QA	Consolidation of metrics and final report

3.4 General Rules

- The supervisor follows each test and makes sure it is **repeated at least three times**.
- Test speeds can be adjusted to the operating requirements of each customer.
- Before each scenario, confirm that the vehicles are registered and communicating.

> ⚠️ **NOTE**: Distances and speeds below are reference values. Agree them with the customer's safety area before starting.

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4. EXECUTION

4.1 V2V: Haul Roads (Straight)

Test L1: Head-on approach (Annex A)

Objective: validate detection and alert progression for a frontal approach. *Equipment:* LO = truck; RO = truck (stationary). *Initial separation:* 100 m. *LO speeds:* 10, 20 and 30 km/h.

Step	Action	Expected Result
4.1.1	Confirm V2V communications	Both vehicles visible to each other
4.1.2	Position LO and RO	Vehicles at the initial separation
4.1.3	LO advances at a stable speed	Progressive alerts to the LO operator before the vehicles meet
4.1.4	Repeat 3 times for each speed	Alert levels and distances recorded per repetition

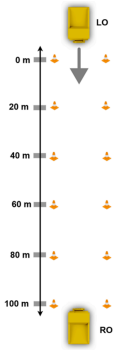


Figure 1: Scenario L1 – Controlled straight approach

Test L2: Reversing (Annex B)

Objective: validate detection of the RO during reversing manoeuvres. *Equipment:* LO = truck (reversing); RO = stationary truck behind. *Initial separation:* 100 m. *LO reversing speeds:* 5 and 10 km/h.

Step	Action	Expected Result
4.1.5	Start recording	Event logging active
4.1.6	LO starts reversing; the operator monitors alerts	Progressive alerts to the operator while approaching the RO

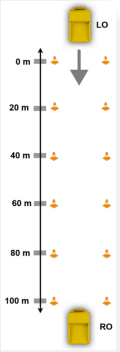


Figure 2: Scenario L2 – Reversing trajectory toward an obstacle

Test L5: Passing head-on (Annex C)

Objective: validate detection of a vehicle crossing on parallel roads in opposite directions. *Equipment:* LO = truck; RO = truck.

Step	Action	Expected Result
4.1.7	LO and RO keep 20 km/h at 100 m frontal separation	Detection and alerts according to the configuration
4.1.8	LO and RO keep 20 km/h at 5 m and 10 m lateral separation	Lateral separation evaluated; alerts consistent with the configured zones

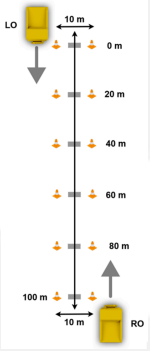


Figure 3: Scenario L5 – Parallel trajectories in opposite directions

4.2 V2V: Curves and Reduced Visibility

Test C1: Head-on curve (Annex E)

Objective: anticipate and alert before direct visibility. *Equipment:* LO = truck; RO = stationary truck. Place the vehicles at the ends of a curve without line of sight. *LO speeds:* 10, 20 and 30 km/h.

Step	Action	Expected Result
4.2.1	Both vehicles advance until they meet; evaluate the first instant of warning	First warning recorded
4.2.2	Record whether the warning occurs before the operators have direct vision	Warning before direct visibility

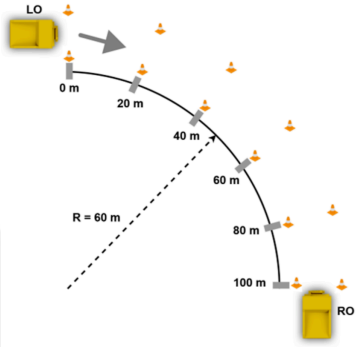


Figure 5: Scenario C1 – Head-on encounter in a curve

4.3 Intersections and Crossings

Test T4: 90° intersection (Annex D)

Objective: validate the crossing-priority calculation and alert prioritisation at a 90° crossing. *Equipment:* LO = crossing vehicle; RO = main road vehicle. Place the vehicles more than 100 m from the crossing. *Speeds:* 10, 20 and 30 km/h.

Step	Action	Expected Result
4.3.1	LO approaches with the intention of crossing; RO keeps its speed	Alerts prioritised according to who has priority
4.3.2	Record the moment of each alert level	Level timings recorded

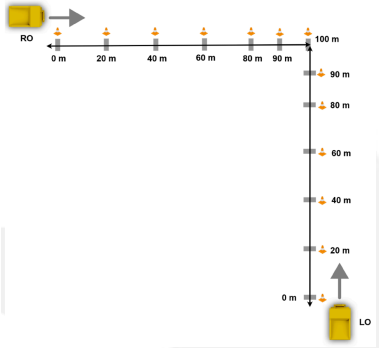


Figure 4: Scenario T4 – Perpendicular crossing between vehicles

Test R2: Cleaning flow, request and authorisation (Annex F)

Objective: validate the digital authorisation for the cleaning tractor.

Step	Action	Expected Result
4.3.3	Tractor (LO) requests permission from the CMS screen	Request reaches the shovel
4.3.4	Shovel operator receives the request and denies it	Request denied

Step	Action	Expected Result
4.3.5	Validate that, without approval, the tractor receives an alert and does not enter	Alert to the tractor
4.3.6	Tractor (LO) requests permission again	Request reaches the shovel
4.3.7	Shovel operator receives the request and approves it	Request approved
4.3.8	Validate that, if the tractor enters the risk zone, the shovel receives an alert	Alert to the shovel operator

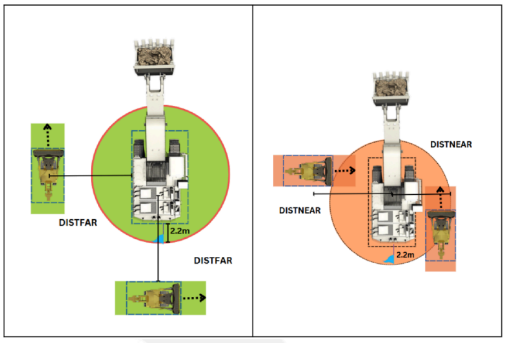


Figure 6: Scenario R2 – Operational cleaning flow

4.4 V2O: Safety Berm (Optional)

Test O1: Reversing toward a berm (Annex G)

Objective: validate distance measurement and alerts when reversing toward a berm. Only where the AI camera add-on is installed. *Speeds:* 5 and 10 km/h, reversing. Three repetitions per speed. Record at cones placed at 15, 10 and 5 m.

Step	Action	Expected Result
4.4.1	Truck reverses toward the berm at the test speed	Berm detected and distance measured
4.4.2	Record the alert at the cones at 15, 10 and 5 m	Progressive alerts as the distance decreases



Figure 7: Scenario O1 – Truck approaching a safety berm

5. VALIDATION

5.1 Performance Criteria (KPIs)

KPI	Target
Effectiveness of V2V alert detection	≥ 95 %
Effectiveness of V2O alert detection (optional module)	≥ 90 %
Alert latency to the operator	≤ 1 s

KPI	Target
Cleaning requests sent successfully	≥ 95 %

⚠️ **NOTE**: KPIs are measured only during functional tests.

5.2 Verification Checklist

	Verification	GO	NO-GO
1	L1: three repetitions per speed with progressive alerts	☐	☐
2	L2: alerts while reversing at 5 and 10 km/h	☐	☐
3	L5: detection at frontal and lateral separations	☐	☐
4	C1: warning before direct visibility	☐	☐
5	T4: alert prioritisation at the crossing	☐	☐
6	R2: denial, approval and risk-zone alerts	☐	☐
7	O1 (if applicable): alerts at 15, 10 and 5 m	☐	☐
8	Events notified to the platform	☐	☐
9	KPIs met	☐	☐

6. IF IT FAILS

Symptom	Probable Cause	Corrective Action	Verification
A vehicle is not seen by the other	Missing registration or dimensions in the local configuration; system or communication fault	Check both configuration entries and the system status messages	Vehicle detected
Registered vehicle seen but no alert	Alert evaluation or configured parameters	Review the evaluation on the display and the parameters with the CMS specialist	Alert generated
Local alerts without records on the platform	Sending or communications to the platform	Check the platform communication	Events visible on the platform
KPI not met	Parameters or positioning conditions	Review the configuration with the CMS specialist and repeat the scenario	KPI met

7. TENTATIVE SCHEDULE

Reference schedule for a Level 8 phase with one shovel, one tractor and two trucks. Durations depend on the fleet and the site.

Activity	Owner	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
Pre-installation reports	Installation team	■	■	■	■						
Support design	Installation team	■	■	■	■						
Material delivery	TORSA project manager	■	■	■	■						
Platform deployment	Infrastructure			■	■						
Pre-configuration	Installation team			■	■						
Shovel installation	Installation team					■					

Activity	Owner	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
Tractor installation	Installation team						■				
Truck installation (x2)	Installation team							■			
Commissioning	CMS specialist					■	■	■			
Validation and phase close (this protocol)	Installation team / CMS specialist								■	■	■

ANNEXES

The following annexes complement this protocol with reference operational diagrams, test layouts and the record format used during field execution.

A. Scenario L1: Straight trajectory

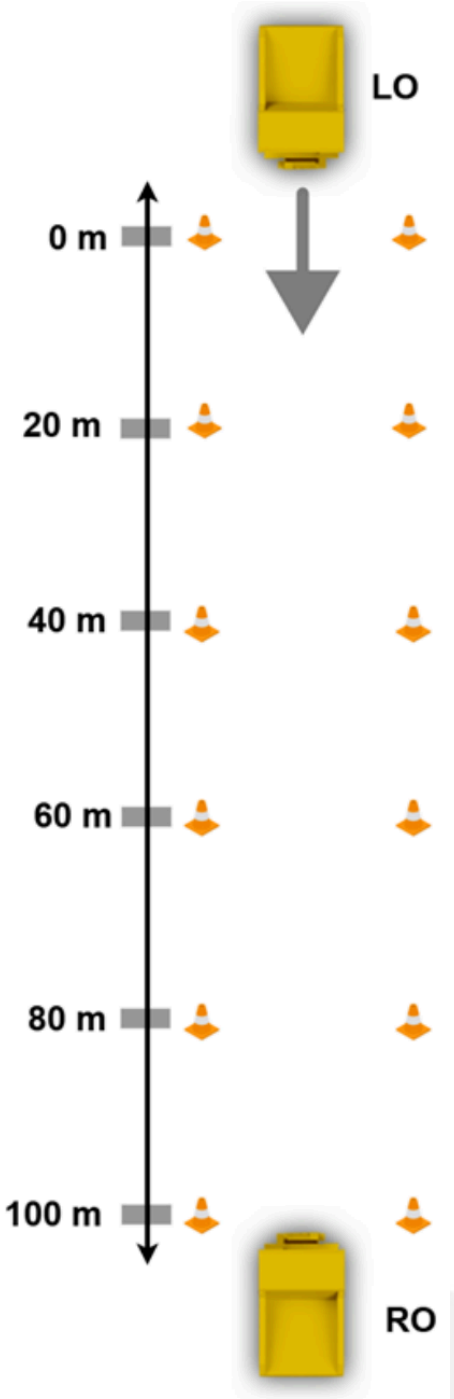


Figure 1: Scenario L1 – Controlled straight approach

B. Scenario L2: Controlled reversing

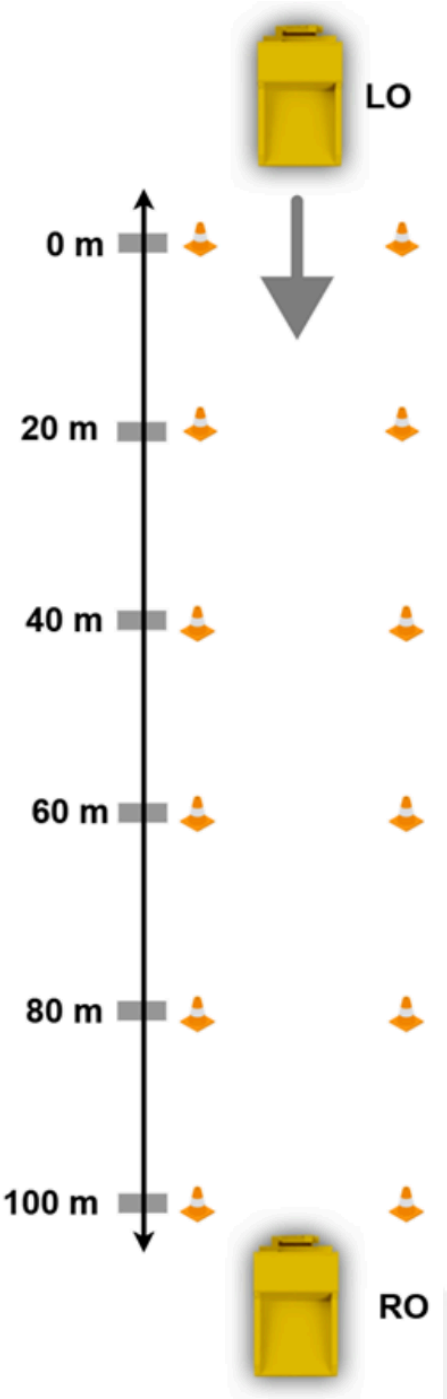


Figure 2: Scenario L2 – Reversing trajectory toward an obstacle

C. Scenario L5: Parallel trajectories in opposite directions

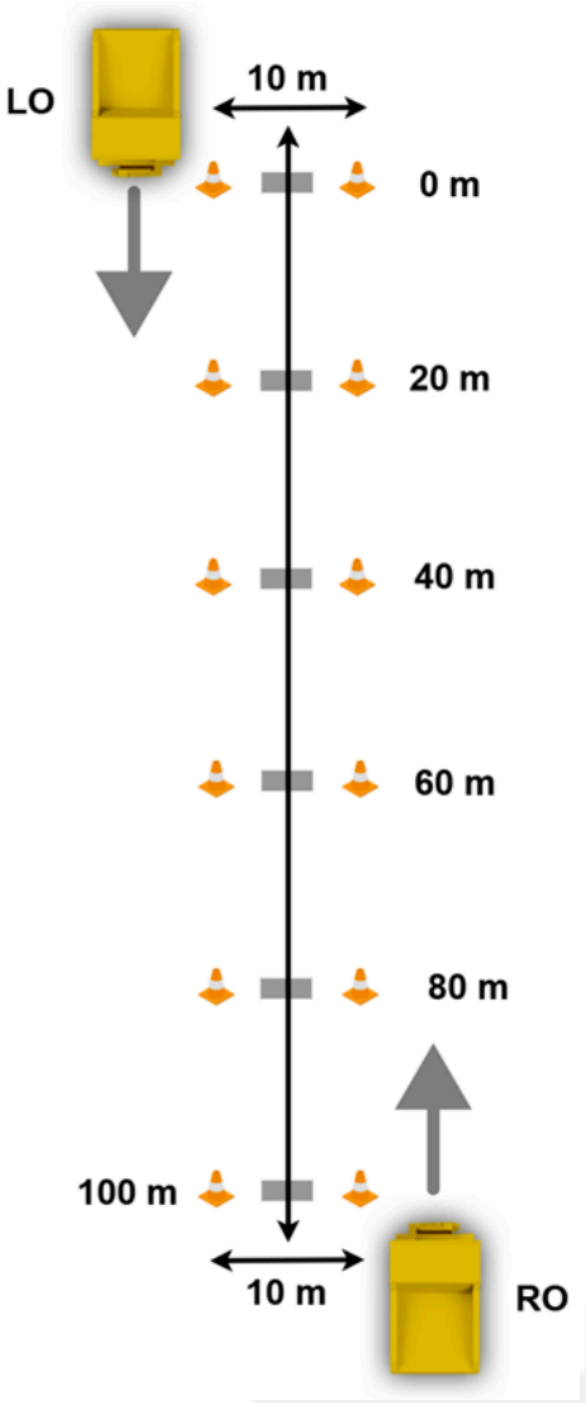


Figure 3: Scenario L5 – Parallel trajectories in opposite directions

D. Scenario T4: Perpendicular intersection

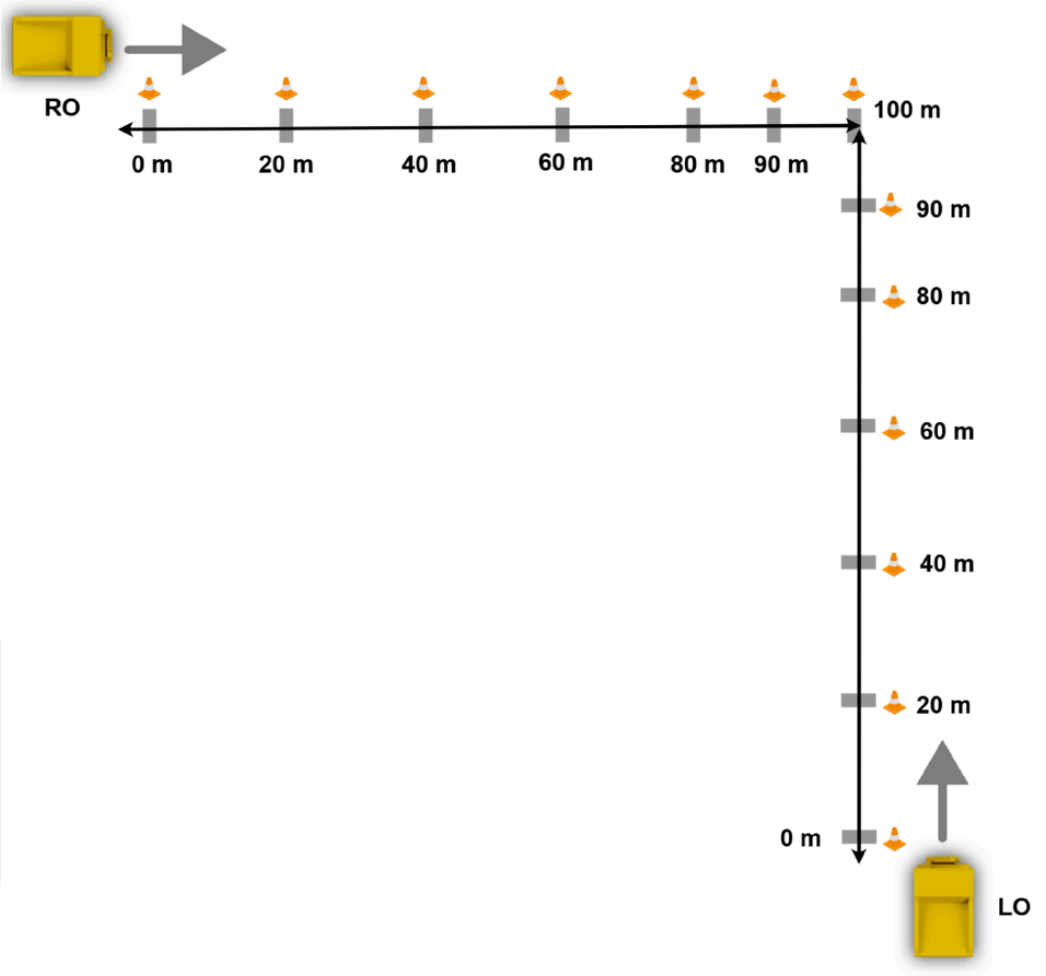


Figure 4: Scenario T4 – Perpendicular crossing between vehicles

E. Scenario C1: Head-on curve

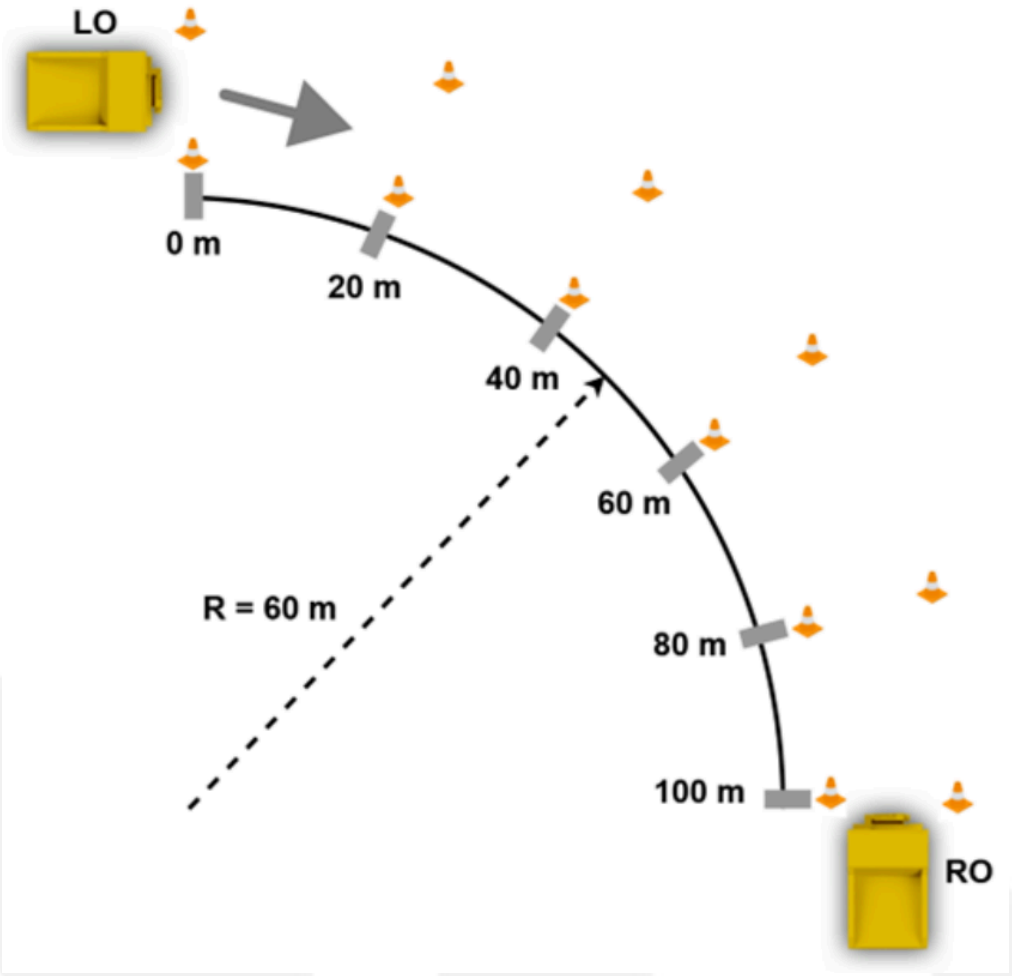


Figure 5: Scenario C1 – Head-on encounter in a curve

F. Scenario R2: Cleaning flow

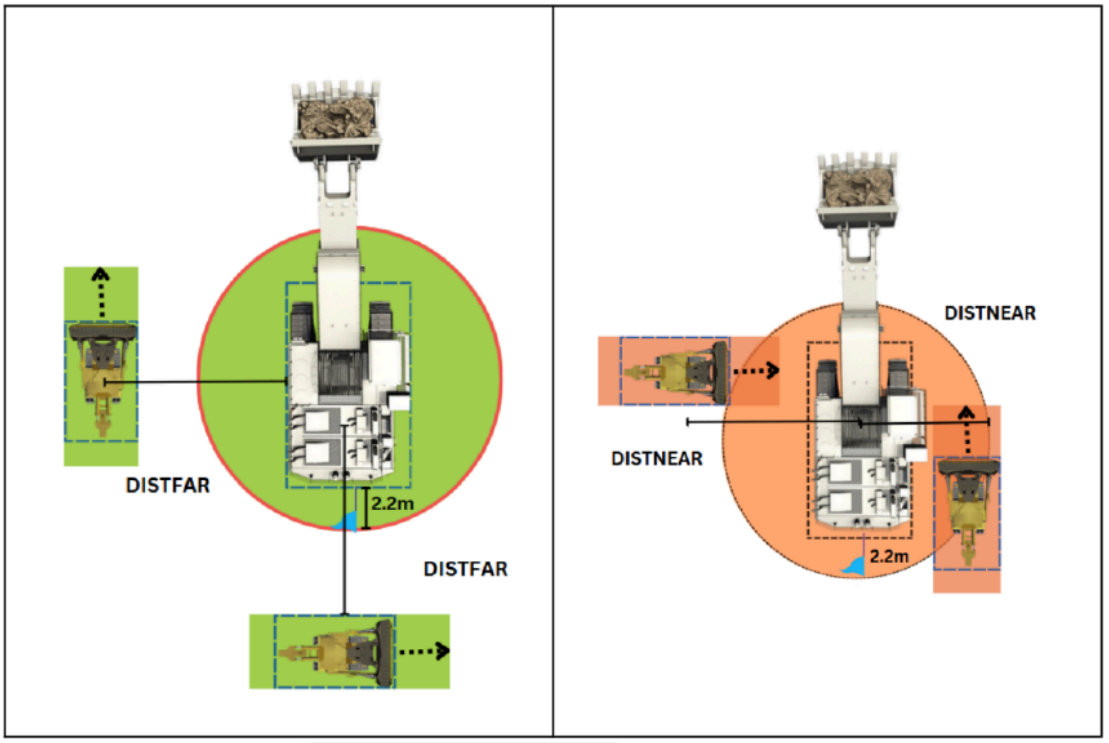


Figure 6: Scenario R2 – Operational cleaning flow

G. Scenario O1: Truck approaching a safety berm (optional V2O)

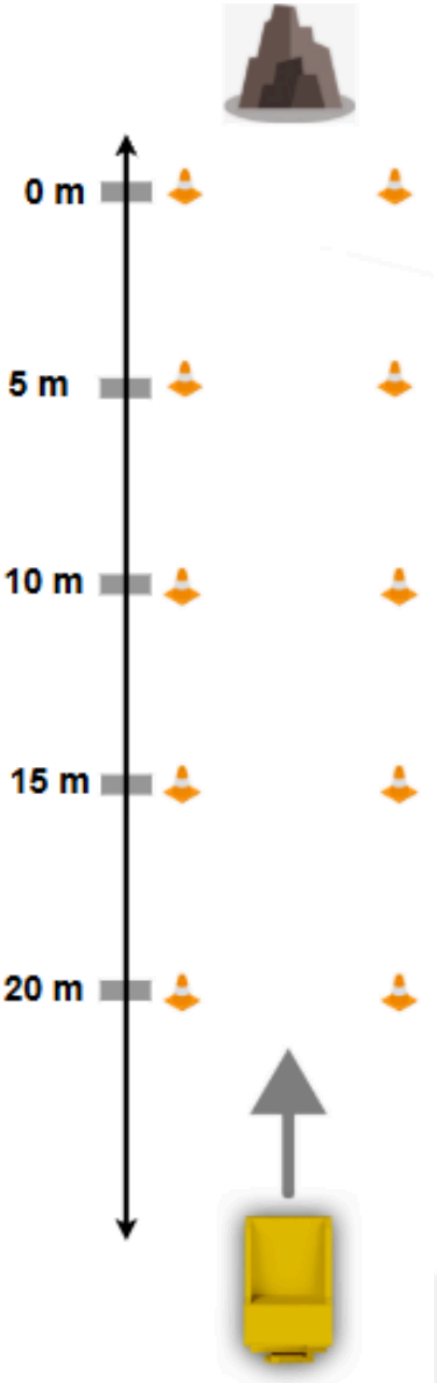


Figure 7: Scenario O1 – Truck approaching a safety berm

ANNEX H: TEST RECORD FORMAT

Scenario	Repetition	LO speed (km/h)	Separation (m)	First alert (m / s)	Alert level reached	Platform notification	Result (GO / NO-GO)
	1					☐	
	2					☐	
	3					☐	

8. REVISION CONTROL

Version	Date	Prepared by	Approved by	Changes
1.0	2026-10-20	TORSA Engineering	—	Generic CMS L8 protocol, anonymised from a customer test protocol; L9 tests removed; tentative schedule added

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TORSA Worldwide S.L. - Severo Ochoa, 19 - Parque Tecnológico de Andalucía - 29590 Málaga, Spain